ELEC60019
ELEC70059

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC
ENGINEERING EXAMINATIONS 2022

MACHINE LEARNING

Friday, 9 December 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer THREE questions. All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Please indicate which 3 questions are to be marked.

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Second Marker(s) : D. Gunduz

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MACHINE LEARNING EXAM PAPER 2022 PLEASE ANSWER 3 OUT OF 4 QUESTIONS.

1. Maximum-Margin Classifiers - Support Vector Machines - Kernel Methods.

a) Consider the following data samples:

$$x_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad x_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \qquad x_3 = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$$

where x_1 and x_2 belong to the negative class, while x_3 belongs to the positive class. Answer the following questions:

i) Draw these points on a Cartesian plane and show that the data is linearly separable. **[5 marks]**

ii) Predict the support vectors by inspecting your plot. **[5 marks]**

iii) Use Lagrangian and dual formulation concepts to obtain the optimal hyperplane equation. (Hint: Find α_i for all the data samples using the dual formulation, where α_i are the Lagrange multipliers corresponding to the inequality constraints.) **[35 marks]**

iv) Do you expect the answers to part ii) and part iii) to be exactly the same? What would be the reason for any discrepancy? Justify your answer. **[5 marks]**

b) Remember that for a function $K : \mathscr{X} \times \mathscr{X} \to \mathbb{R}$ to be a valid kernel, it needs to satisfy the Mercer conditions:

- Symmetry, i.e., $K(\mathbf{x},\mathbf{y}) = K(\mathbf{y},\mathbf{x})$.
- Positive semidefiniteness, i.e., for any $n \in \mathbb{Z}^+$ and $\{\mathbf{x}_1,\ldots,\mathbf{x}_n\} \subset \mathscr{X}$, the kernel matrix $K(\mathbf{x}_1,\ldots,\mathbf{x}_n) = (K(\mathbf{x}_i,\mathbf{x}_j))_{ij} \in \mathbb{R}^{n\times n}$ is positive semidefinite. (An $n \times n$ symmetric real matrix A is said to be positive semidefinite if $\mathbf{x}^\mathsf{T} A\mathbf{x} \geq 0$, $\forall \mathbf{x} \in \mathbb{R}^n$.)

Show that the following functions are valid kernels for $\mathbf{x},\mathbf{y} \in \mathscr{X} = \mathbb{R}^d$ and $a,b \in \mathbb{R}^+$.

i) $K_1(\mathbf{x},\mathbf{y}) = \begin{cases} a, & \text{if } \mathbf{x} = \mathbf{y} \\ 0, & \text{otherwise} \end{cases}$. **[10 marks]**

ii) $K_2(\mathbf{x},\mathbf{y}) = \mathbf{x}^T\mathbf{y}$. **[20 marks]**

iii) $K_3(\mathbf{x},\mathbf{y}) = a\mathbf{x}^T\mathbf{y} + b$. **[20 marks]**

2. k-Means Clustering, Principal Component Analysis

a) Consider the following six data samples in $\mathbb{R}^2$:

$$(-1,-2),\ (0,-1),\ (0,0),\ (2,0),\ (3,0),\ (4,1)$$

i) Perform k-means clustering on the data points above for $k = 2$ and Euclidean distance. Use $(0,0)$ and $(4,1)$ as the initial cluster centers. Find the cluster centers and indicate which points belong to each cluster. **[15 marks]**

ii) Consider n data points $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^d$. What is the point in $\mathbb{R}^d$ that minimizes the total squared Euclidean distance to these points? That is, find the solution of the following optimization problem

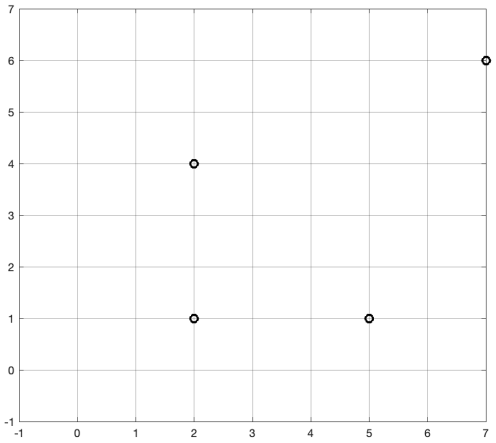
$$\operatorname*{argmin}_{\mathbf{y}\in\mathbb{R}^d} \sum_{i=1}^{n} d(\mathbf{x}_i, \mathbf{y})^2, \tag{2.1}$$

where $d(\mathbf{x}, \mathbf{y})$ is the Euclidean distance between points $\mathbf{x}$ and $\mathbf{y}$. Prove your answer. **[20 marks]**

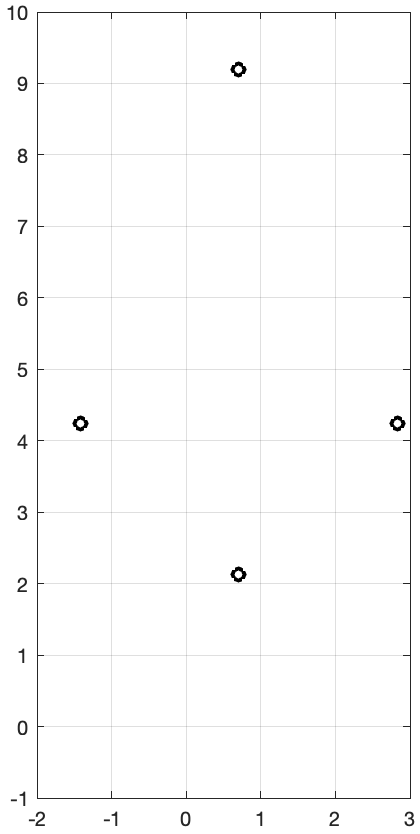
b) Consider the data samples $(2,1), (2,4), (5,1), (7,6) \in \mathbb{R}^2$.

i) Apply the PCA algorithm on these data points and find the unit-length principal component directions. **[30 marks]**

ii) We would like to represent each sample point of the above dataset with a one dimensional representation. Draw a line along the principal component direction on the plot below, project each data point onto this line, and find the principal coordinate value of each data point. **[15 marks]**



iii) We rotate all the data points 45 degrees counterclockwise around the origin. What are the principal component directions for this new dataset? What are the principal coordinate values for the data samples of this new dataset? Draw it on a plot. **[20 marks]**



3. **Linear Regression**: You are building a linear regressor to fit point-label pairs $(x_1, y_1) = (1, 5)$ and $(x_2, y_2) = (2, 7)$ using the linear equation $\hat{y} = mx + b$. Answer the following questions:

a) Calculate the coefficients (weights) (m, b) of the line that fits the data points using the linear equation given above. Plot the points and the line on the Cartesian plane. **[10 marks]**

b) Find the parameters of the linear regressor using **gradient descent**.

- Write down the gradient and the update rule for each coefficient m, b by considering the mean squared error $MSE = \frac{1}{n} \cdot \sum_{i=1}^{i=n} (y_i - \hat{y}_i)^2$ as the loss function. (Hint: include a learning rate α in the update rule) **[10 marks]**
- Write down two iterations of gradient descent assuming a learning rate $\alpha = 0.2$. Set the initial values for (m, b) to $(9, 10)$. For each iteration, including the initialisation, plot the line for (m, n) values against the data points and calculate their MSE. **[20 marks]**

c) Use the closed form solution approach to calculate the coefficients of a linear regressor. Hint: if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. **[15 marks]**

d) Calculate the coefficients using ridge regression approach with $\lambda = 1$. **[15 marks]**

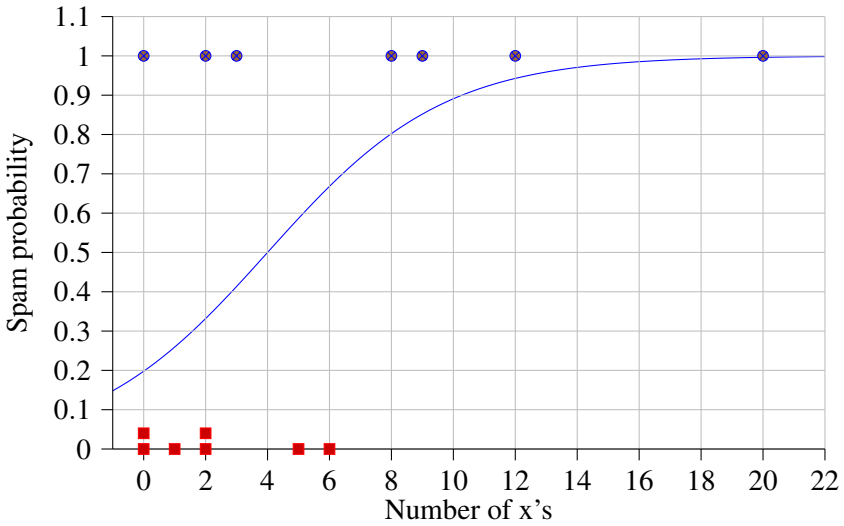
e) Apply non-linear transformation to the original data points using the the following function $\phi(x) = (x, 3x - 1, 2x^2 + 2)$ and discuss its usefulness. Propose better transformations A, B, C for $\phi(x) = (A, B, C)$ and explain why they are better. **[15 marks]**

f) Design a neural network with 2 layers that can perform multivariate regression of 2 variables for 3-dimensional input. Give as many details as possible and justify briefly your design choices. Draw a diagram of this network. **[15 marks]**

4. **Logistic Regression**: Suppose that you are building a spam filter. You develop a predictor based on the occurrence of letter 'x' as an indicator for the email to be spam or non-spam. There are 7 spam messages and 7 non-spam messages represented by the count of letter 'x' in each message.

- Occurrence of letter 'x' in 7 spam messages: [0, 2, 3, 8, 9, 12, 20]
- Occurrence of letter 'x' in 7 non-spam messages: [0, 0, 1, 2, 2, 5, 6]

You have trained a logistic regression classifier with spam messages in class 1 and non-spam messages in class 0. The data points and the logistic function are plotted below.



Answer the following questions:

a) What is the probability (approximate) of a message without 'x' to be a spam and a non-spam according to the trained model and data points? **[10 marks]**

b) How many letters 'x' must a message contain to be classified as a spam with 100% certainty? **[10 marks]**

c) How can we make binary classification decisions using this logistic regression model that would minimise the classification error on the data given above? **[10 marks]**

d) Using the approach you proposed in c):

- apply it to the data points and write down the resulting confusion matrix reporting true positives, true negatives, false positives and false negatives,
- calculate the total accuracy.

[20 marks]

e) Assume that the cost of classifying a non-spam as a spam is very high. How

can the goal of minimizing the cost be defined in terms of precision and recall? **[10 marks]**

f) Suppose that each data point was extended with a random feature, which increased the dimensionality of the data. Discuss how this action would affect the complexity, training error, test error, variance, bias, and overfitting of the model. **[10 marks]**

g) Spam data points were extended to 2 dimensional vectors $\mathbf{x}$ by including another feature from the emails. The points are illustrated below with $+$ and $-$ indicating spam and non-spam, respectively. There are two perceptrons $\mathbf{w}_A$ and $\mathbf{w}_B$, each having three parameters $\mathbf{w} = (w_0, w_1, w_2)$. These two perceptrons are included in a Multi Layer Perceptron (see diagram) that correctly classifies all the points. Use the figures below to find the values of the parameters $\mathbf{w}_A$, $\mathbf{w}_B$ and $\mathbf{w}_C$. **[30 marks]**

